UML Component Diagram

Peer Skill Exchange & Mentorship Network - Simple Layered Architecture

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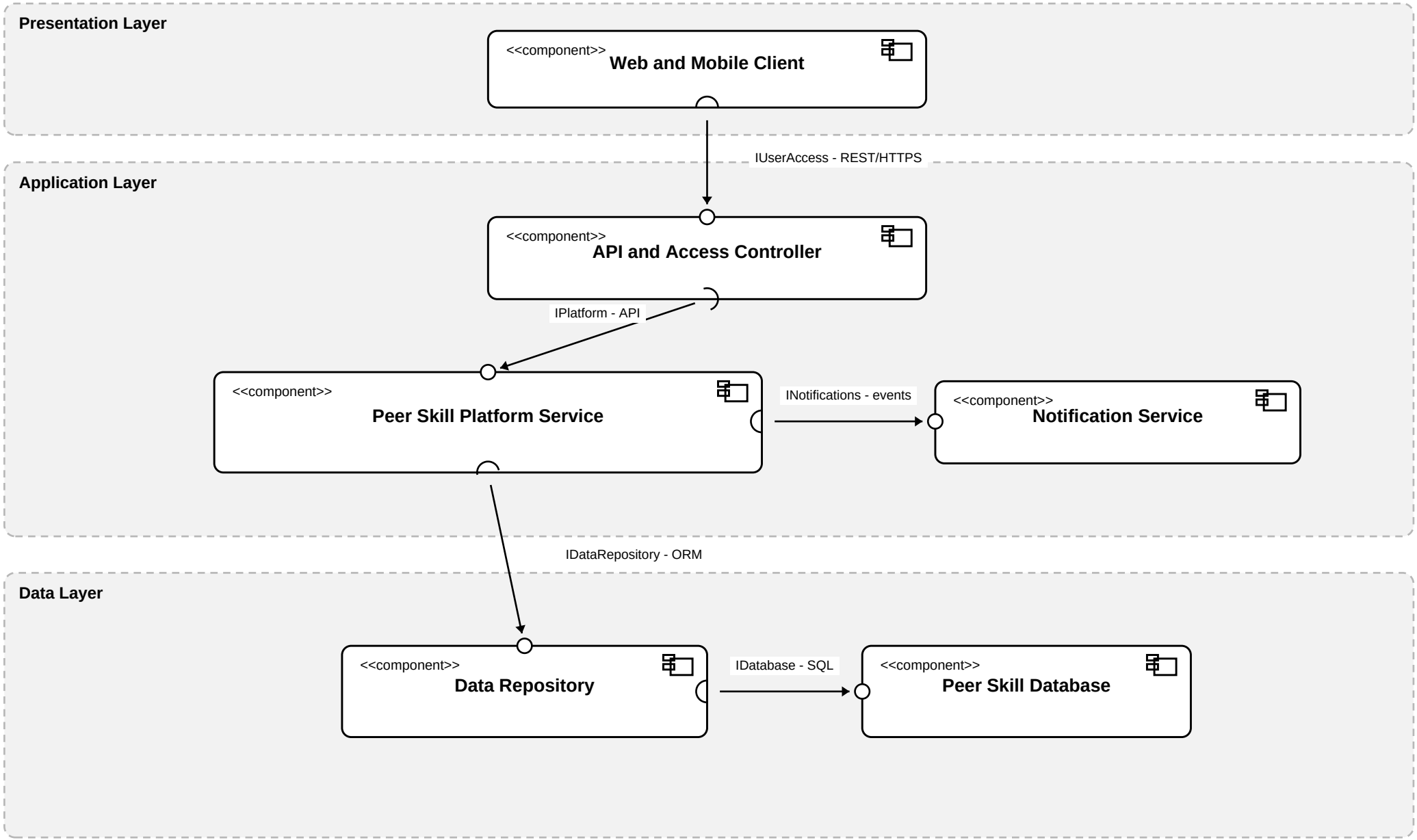

Legend: provided interface | required interface | request or data flow

6 components | 5 labeled interfaces

Architecture Justification

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Architecture Choice

We chose a layered architecture. The client handles the interface, the application layer handles system rules, and the data layer stores information.

Why It Fits

First, each layer has one clear job, so changes are easier to make and test. Second, this design is simple enough for a small project and avoids the extra deployment work of microservices.

Security Advantage

Every request passes through the API and Access Controller, which checks login and permission. The database is not exposed directly to users.

Performance Benefit

Database indexes and a small cache can speed up member searches. Keeping the main features in one application also avoids extra network calls.